



**African Population and
Health Research Center**

Transforming lives in Africa through research.

ASSESSING THE DATA QUALITY OF OMOP CDM USING DATA QUALITY DASHBOARD AND ACHILLES

Reinpeter Momanyi
18th September 2026

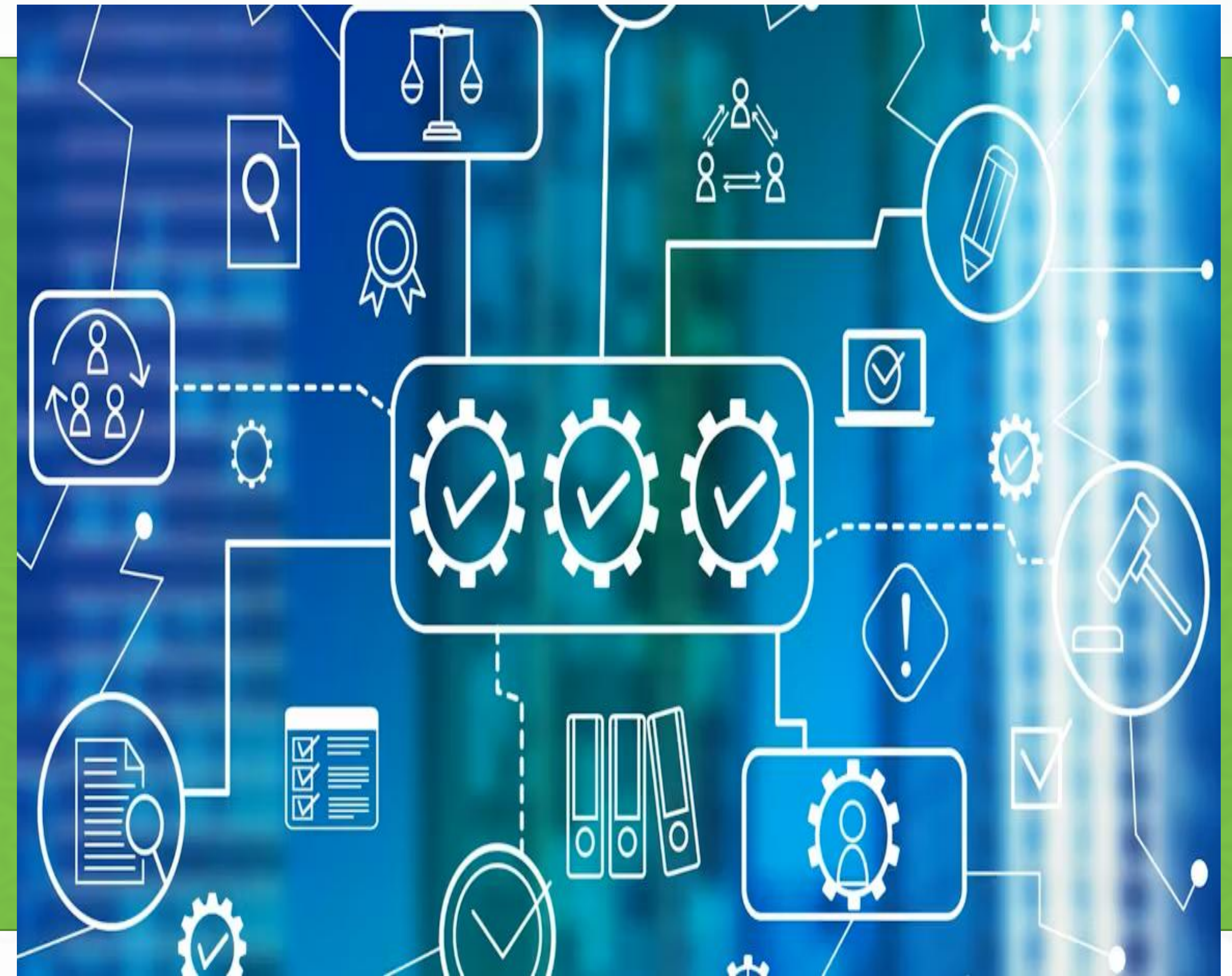


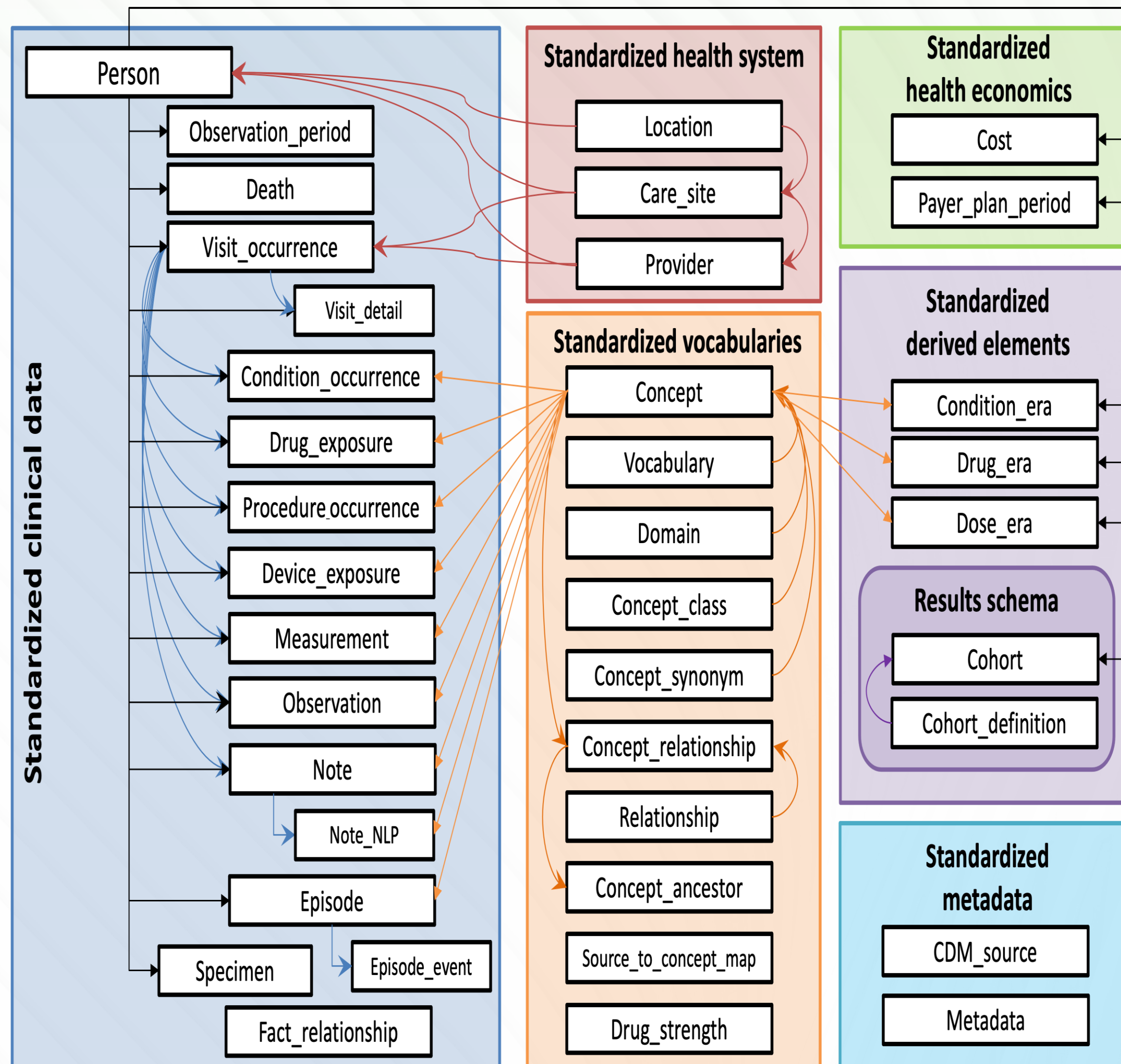
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1. Overview of OMOP Common Data Model (CDM)



Structure of CDM v5.4

39 Tables

27 to standardize data and metadata; 10 for standardized vocabularies; 2 for results

Aim

Bringing data into a common format that allows for collaborative research, large-scale analytics, and the sharing of sophisticated tools and methodologies.

Standardized vocabularies allow the organization and standardization of medical terms to be used across the various clinical domains

Why?

- ✓ Evidence generation using standardized open-source analytics tools
- ✓ Data quality and characterization (i.e. counting occurrences of events)
- ✓ Patient-level predictive modelling
- ✓ Medical product safety surveillance

2. Data Quality

- ✓ An important aspect of transforming source data to OMOP-CDM deals with assessing the data quality for observational healthcare research purposes.
- ✓ If the quality of the data is poor (garbage in), then the quality of the result of research using that data must be poor as well (garbage out).

What is data quality?

- ✓ The state of completeness, validity, consistency, timeliness and accuracy that makes data appropriate for a specific use.
 - **Completeness:** availability of the minimum set of attributes required by the End User to meet objectives
 - **Accuracy:** correctness of the data
 - **Consistency:** uniformity of data across different systems
 - **Validity:** adherence to the defined set of rules & standards
 - **Uniqueness:** redundancy of the same record in the data
 - **Timeliness:** availability of data and how current/updated it is



**GARBAGE IN,
GARBAGE OUT**



If the quality of the data is poor, then the quality of the result of research using that data must be poor as well.

Causes of poor Data Quality



This is one of the most common causes where lack of training to use tools, motivation and other distractions at work result in typos, missing details, data entry in wrong fields etc.

Human Errors



This issue happens usually while updating legacy systems and may result in data quality issues due to mapping errors, processing errors or duplicate check errors.

Data Migration & Conversion



Integration Issues



Poor or lack of integration between systems, results in data silos and unavailability of valuable data at the point of business consumption.

Outdated Data

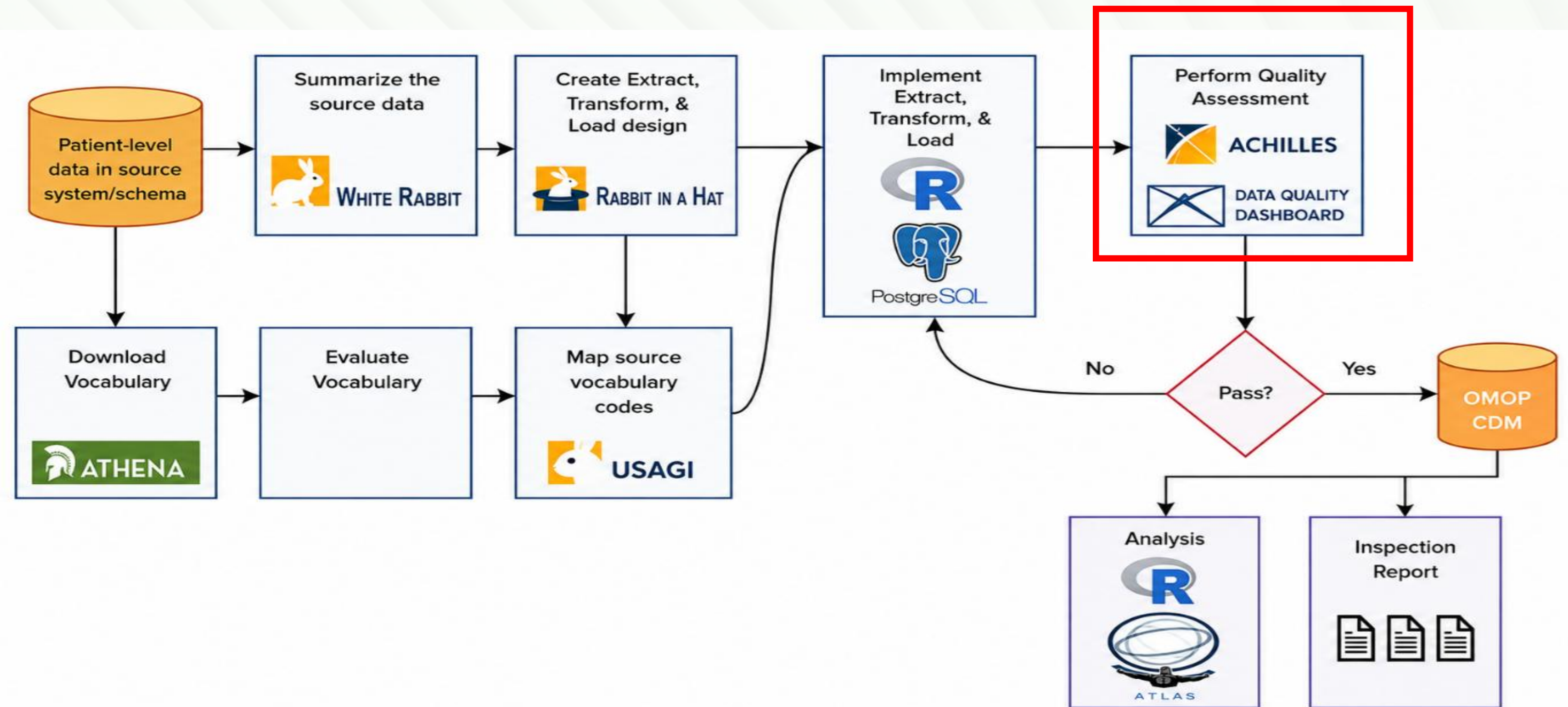


Data is time sensitive. Changing addresses, contact details of customers. Changing product features with time. If not updated results in poor business outcomes.

Transformational Errors



These are the errors caused while transforming data from one form to the other to meet the different system reqmnts. Eg- dates, Phone numbers, zip codes etc.



- ✓ OHDSI has developed 2 tools to assist in the assessment of data quality against the OMOP-CDM data before using the data for research
 - Data Quality Dashboard
 - Achilles
- ✓ Before the execution of DataQualityDashboard and Achilles, it is required that a **results schema** be created.
- ✓ Data Quality Dashboard is an Iterative process



3. Data Quality Dashboard (DQD)

- ✓ This R package runs a series of data quality checks against an OMOP CDM instance (currently supports v5.4, v5.3, and v5.2 but only tested against v5.3 and v5.4)
- ✓ Evaluates the checks against some pre-specified threshold i.e Does a **table check, field check, or concept-level check**
- ✓ Produces a set of data check results with supplemental investigation assets
- ✓ Quality checks are organized according to the **Kahn Framework**
 1. **Conformance:** adherence to specified standards and formats
 - **Value:** Are recorded data elements in agreement with the specified formats?
 - **Relational:** Is recorded data in agreement with specified relational constraints?
 - **Computation:** Do computations on the data yield the intended results?
 2. **Completeness:** whether a particular variable is present and contains all recorded values e.g. do all persons have a known gender?
 3. **Plausibility:** Are data values believable
 - **Uniqueness:** E.g., does each PERSON_ID occur only once in the PERSON table?
 - **Atemporal:** Do values, distributions, or densities agree with expected values?
 - **Temporal:** Are changes in values in line with expectations?
- ✓ Before execution, the **DataQualityDashboard package** requires that the **CDM_SOURCE table** be properly populated



Data Quality Check Types



Check Type	Check Description
Person Completeness	The number and percent of persons in a database that do not have a least one record in the <i>CDM table</i> .
Is Required	The number and percent of records with a NULL value in a <i>CDM field</i> of a <i>CDM table</i> that is considered not nullable.
Is Primary Key	The number and percent of records that have a duplicate value in the <i>CDM field</i> of the <i>CDM table</i> .
Is Foreign Key	The number and percent of records that have a value in a <i>lookup field</i> of a <i>CDM table</i> that does not exist in the <i>lookup table</i> .
Concept Domain	The number and percent of records that have a <i>concept_id</i> that does not conform to the <i>domain</i> .

Check Type	Check Description
Is Standard Valid Concept	The number and percent of records that do not have a standard, valid concept in the <i>CDM field</i> of a <i>CDM table</i> .
Standard Concept Completeness	The number and percent of records with a value of 0 in the standard concept field <i>CDM field</i> in the <i>CDM table</i> .
Plausible Temporal After	The number and percent of records with a value in a <i>CDM field</i> of a <i>CDM table</i> that occurs prior to a <i>plausible date</i> .
Plausible Value Low	For a given <i>concept_id</i> and <i>unit_concept_id</i> pair, the number and percent of records with a value lower than the <i>plausible low value</i> .
Plausible Gender	For a given <i>concept_id</i> , the number and percent of records associated with persons with an <i>implausible gender</i> .

4. DQD Demo

5. Achilles (Automated Characterization of Health Information at Large-scale Longitudinal Evidence Systems)



- ✓ This R package provides characterization against an OMOP CDM instance (currently supports v5.4, v5.3)
- ✓ Each data characterization analysis has an analysis ID and a short description of the analysis
- ✓ Produces a set of results with descriptive statistics stored in a database.
- ✓ The stored results can be integrated into ATLAS (a web-based tool) as the “Data Sources” function that enables visualization access.



A vibrant outdoor market scene featuring several women. In the foreground, a woman in a yellow and red patterned dress is smiling and selecting from a large pile of ripe red tomatoes. To her right, another woman in a blue and white patterned dress is also smiling. In the background, other women and market stalls with various goods are visible. The scene is set in a bustling, open-air market environment.

6. Achilles Demo





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